

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack Development Using MERN
Practical – II

Roll No.:T006	Name:Nandini Chaudhari
Class:TYIT	Batch:01
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Aim-2. Core JavaScript Concepts for MERN

A. Create functions and call them with parameters.

CalculateSalary.

Code:-

```
function calculateSalary(name,basicSalary) {
  let bonus = 5000;
  let totalSalary = basicSalary + bonus;
  console.log("Employee Name:",name);
  console.log("Basic Salary:",basicSalary);
  console.log("Bonus:",bonus);
  console.log("Total Salary:",totalSalary);
}
calculateSalary("Nandini", 40000)
```

Output:-

```
PS D:\MERN PRACTICAL> node prac2a.js
Employee Name: Nandini
Basic Salary: 40000
Bonus: 5000
Total Salary: 45000
```

GenerateBill

Code:-

```
function generateBill(product,quantity,price) {
  let total = quantity * price;
  let bill = total + (total * 0.09);
  console.log("Product Name:",product);
  console.log("Product Quantity:",quantity);
  console.log("price per Item:",price);
  console.log("Total Bill:",bill);
}
generateBill("TV",1,40000)
console.log("Nandini")
```

Output:-

```

PS D:\MERN PRACTICAL> node practical1a1.js
Product Name: TV
Product Quantity: 1
price per Item: 40000
Total Bill: 43600
Nandini

```

B. Work with arrays and objects in JavaScript.**Code:-**

```

let fruits = ["mango","apple","banana","cherry"];
let student = {
  name:"Nandini",
  age:20,
  cty:"Mumbai"
};
console.log("Fruits:",fruits);
console.log("Seconf Fruit:",fruits[1]);
console.log("Student Name:",student.name);
console.log("Student Age:",student.age);
console.log("Student City:",student.cty);
console.log("Nandini");

```

Output:-

```

PS D:\MERN PRACTICAL> node prac2b.js
Fruits: [ 'mango', 'apple', 'banana', 'cherry' ]
Seconf Fruit: apple
Student Name: Nandini
Student Age: 20
Student City: Mumbai
Nandini

```

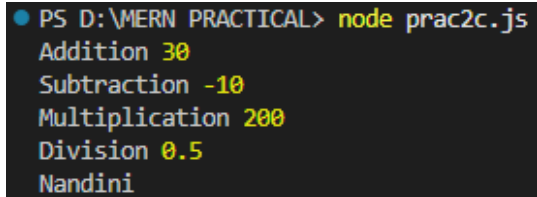
C. Demonstrate use of arrow functions.**Code:-**

```

let add = (a,b) => {
  return a + b;
};
let sub = (a,b) => {
  return a - b;
};
let mul = (a,b) => {
  return a * b;
};
let div = (a,b) => {
  return a / b;
};
console.log("Addition",add(10,20));
console.log("Subtraction",sub(10,20));

```

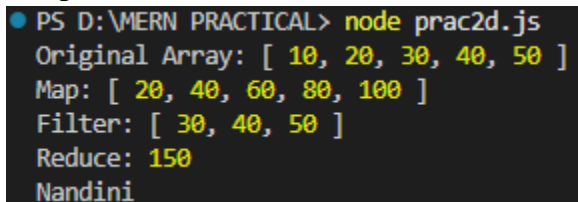
```
console.log("Multiplication",mul(10,20));
console.log("Division",div(10,20));
console.log("Nandini");
```

Output:-


```
PS D:\MERN PRACTICAL> node prac2c.js
Addition 30
Subtraction -10
Multiplication 200
Division 0.5
Nandini
```

D. Write a program using array methods (map, filter, reduce).**Code:-**

```
let numbers = [10, 20, 30, 40,,50];
//map()
let doubled = numbers.map(num => num * 2);
// filter()
let greater25 = numbers.filter(num => num > 25);
// reduce()
let total = numbers.reduce((sum, num) => sum + num, 0);
console.log("Original Array:", numbers);
console.log("Map:", doubled);
console.log("Filter:", greater25);
console.log("Reduce:", total);
console.log("Nandini");
```

Output:-


```
PS D:\MERN PRACTICAL> node prac2d.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
Nandini
```

E. Create a program that manipulates objects and displays output.**Code:-**

```
let student = {
  roll_No : 6,
  name: "Nandini",
  marks: 85
};
console.log("Original Object: ");
console.log(student);
console.log("\nAccessing Properties: ");
console.log("Name: ",student.name);
console.log("Marks: ",student.marks);
student.marks = 90;
```

```

console.log("\n After Updating Marks: ");
console.log(student);
student.city = "Mumbai";
console.log("\n After Adding City: ");
console.log(student);
delete student.roll_No;
console.log("\n After Deleting Roll Number: ");
console.log(student);
console.log("Nandini")

```

Output:-

```

PS D:\MIERN PRACTICAL> node prac2e.js
Original Object:
{ roll_No: 6, name: 'Nandini', marks: 85 }

Accessing Properties:
Name: Nandini

Accessing Properties:
Name: Nandini
Marks: 85

After Updating Marks:
{ roll_No: 6, name: 'Nandini', marks: 90 }

After Adding City:
{ roll_No: 6, name: 'Nandini', marks: 90, city: 'Mumbai' }

After Deleting Roll Number:
{ name: 'Nandini', marks: 90, city: 'Mumbai' }
Nandini

```

F. Implement a small JavaScript program combining functions, arrays, and objects.

Code:

```

let students = [
  {
    roll_No: 100,
    name: "Rahul",
    marks: 85
  },
  {
    roll_No: 102,
    name: "Varun",
    marks: 85
  },
  {
    roll_No: 103,
    name: "Anju",
    marks: 90
  }
]

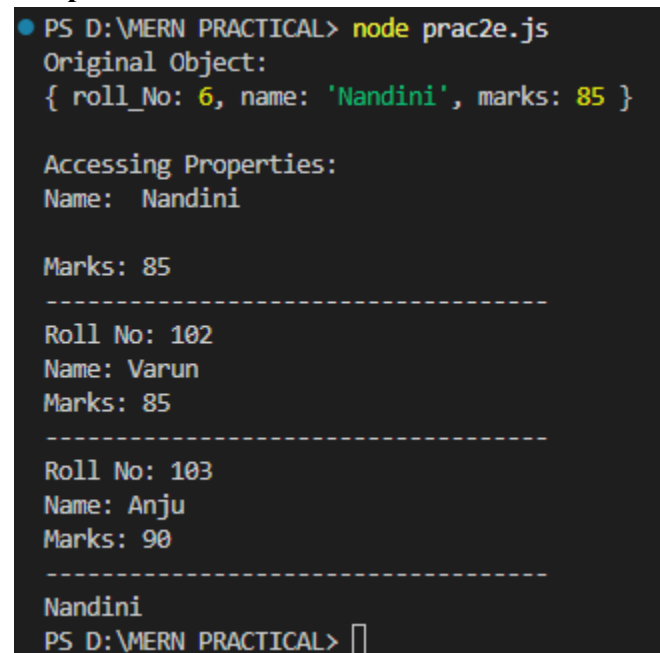
```

```

}
];
function display(s)
{
  console.log("Student Details");
  console.log("-----");
  for(let stud of s)
  {
    console.log(`Roll No: ${stud.roll_No}`);
    console.log(`Name: ${stud.name}`);
    console.log(`Marks: ${stud.marks}`);
    console.log("-----");
  }
}
display(students);
console.log("Nandini")

```

Output:-



```

PS D:\MERN PRACTICAL> node prac2e.js
Original Object:
{ roll_No: 6, name: 'Nandini', marks: 85 }

Accessing Properties:
Name:  Nandini

Marks: 85
-----
Roll No: 102
Name:  Varun
Marks: 85
-----
Roll No: 103
Name:  Anju
Marks: 90
-----
Nandini
PS D:\MERN PRACTICAL> 

```